MIL-H-60650A(MU)
16 February 1971
SUPERSEDING
MIL-H-60650(MU)
6 September 1967

MILITARY SPECIFICATION

HEAD ASSEMBLY, TANK PERISCOPE: 10542204

1. SCOPE

1.1 This specification covers one type of periscope head assembly which deflects a line of sight to the body assembly of a tank periscope. A rotatable mirror provides for elevating and depressing the field of view (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester - Two Station
F10542204	Head Assembly, 10542204
F10513500	Mount Assembly

FSC 1240

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F11738174	Gage - Interchangeability - Body Maximum
F11738175	Gage - Interchangeability - Body Minimum
F11738178	Gage - Interchangeability - Mount Maximum
F11738179	Gage - Interchangeability - Mount Minimum
F11738182	Fixture - Pressure Testing - Body Maximum
F11738183	Fixture - Pressure Testing - Body Minimum
F11738184	Fixture - Pressure Testing - Mount Nominal

PACKAGING DATA SHEET

10542204	Packaging of Head Assembly: 10542204

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The head assembly shall be manufactured in accordance wit Drawing F10542204 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with the requirements of MIL-F-13926 specified below.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The head assembly shall show no evidence of physical failure such as glass breakage and bonding or casting failure due to storage at temperatures ranging from -80° to +160° Fahrenheit (F). Subsequent to meeting this requirement, the head assembly shall be capable of meeting the requirements of 3.4.

3.3.2 Extreme operating temperatures.- The head assembly shall meet the applicable requirements of 3.11 while exposed to and thermally stabilized at temperatures of -65° and +150°F.

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3.3.3 Standard temperature.- Unless otherwise specified herein, the head assembly shall meet the following requirements while exposed to a temperature between 60° and 90°F.

3.4 Vibration. - This requirement shall be met subsequent to meeting the requirements of 3.7.1.1. The head assembly shall show no evidence of physical failure due to being vibrated successively along 3 mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 cycles per second (cps) in 5 minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 (1/32 inch total excursion). Following the vibration test, the head assembly shall meet the requirements of 3.7.1.2, and shall then meet all of the following requirements.

3.5 Cleanliness. - There shall be no visible evidence of condensates, grease, fingerprints and fractures when viewing the glass components by transmitted light from either end. In addition, the total number of dirt (or foreign) particles on the glass surfaces (combined) shall not exceed 4 when viewing from either end. The maximum dimension of any particle shall be 0.025 inch.

3.6 Sealing. - The head assembly, with shipping covers installed, shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) for a minimum of 45 minutes with no evidence of leakage.

3.7 Orientation. - The requirements of 3.7.1 to 3.7.7 inclusive shall be met with the head assembly positioned as follows:

a. Datum Plane “A”, shown on Drawing F10542O4, on a horizontal plane.

b. Surface “c” of the input coupling, shown on Drawing FI0542204, positioned 66° with respect to Datum Plane “A” unless otherwise specified.

3.7.1 Collimation.

3.7.1.1 Pre-vibration. - This requirement shall be met prior to meeting the requirements of 3.4. An incident line of sight which is originally perpendicular to Datum Plane “B”, shown on Drawing F10542O4, shall emerge parallel to Datum Plane “B” within 0.20 mil and perpendicular to the intersection of Datum Planes “A” and “B” within 0.60 mil.

3.7.1.2 Post-vibration. - This requirement shall be met subsequent to meeting the requirements of 3.4. the incident line of sight specified in 3.7.1.1 shall emerge within 0.10 mil of the collimation readings established in 3.7.1.1.

3.7.2 Collimation change (due to assembling to mount). - An original line of sight shall not deviate more than 0.10 mil due to repeated assembling of the head assembly to the mount with which the head assembly is intended to be used.

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3.7.3 Image tilt. - The image of a target line originally perpendicular to Datum Plane "A" shall emerge perpendicular to Datum Plane "B" within 30 minutes of arc.

3.7.4 Excursion range.- Starting with the coupling positioned 67 degrees 30 minutes with respect to Datum Plane "A" rotation of the input coupling shall cause the line of sight to elevate a minimum of 22 degrees and depress a minimum of 18 degrees.

NOTE: At no time shall the input coupling be allowed to strike hard against the travel stops.

3.7.5 Plumb travel. - Starting with the input coupling positioned at the angular value specified in 3.7.4 and the line of sight in coincidence with a target line which is perpendicular to Datum Plane "A", the line of sight shall track the target line within 0.25 mil when elevated and depressed through the range specified in 3.7.4.

3.7.6 Elevation accuracy. - This requirement shall be met with backlash removed and the angle measured in a plane which is perpendicular to Datum Planes "A" and "B". The angle formed by the line of sight and Surface "C" of the input coupling shall remain constant within 0.20 mil at any elevation setting through the range specified in 3.7.4.

3.7.7 Backlash. - Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.8 Resolution.- Resolution shall be 2.0 seconds of arc or less when observing with a telescope of at least 90 power. The entrance pupil of the telescope shall be not less than 7.0 inches. Resolution shall be observed at one focus setting of the telescope.

3.9 Luminous transmittance.- Luminous transmittance shall not be less than 90 percent. The source of radiance shall be at the exit aperture of the head assembly and the observation made at the entrance aperture.

3.10 Coupling travel. - Rotation of the input coupling shall be limited by contact with the travel stop pin only.

3.11 Torque. - The running torque required to rotate the input coupling through the range specified in 3.7.4 shall be within the limits of 3.0 and 7.0 inch-pounds at standard temperature. When checked at the temperatures specified in 3.3.2, the running torque shall not exceed 11.0 inch-pounds.

3.12 Interior atmosphere.- The interior atmosphere of the accepted head assembly, with shipping cover installed, shall contain nitrogen having a dew point at least as low as -25°F and the internal pressure of the head assembly shall be 3 psig ± 0.5 psig.

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3.13 Cleanliness. - With shipping covers installed, there shall be no visible evidence of condensates or fractures when viewing the glass components by reflected light at the front window of the head assembly, the total number of dirt (or foreign) particles on the glass surfaces (combined) shall not exceed 4 when viewing from either end. The maximum dimension of any particle shall be 0.025 inches.

3.14 Interchangeability.

3.14.1 Interchangeability - Mechanical.- The Head Assembly 10558203 shall be capable of accepting all Mount Assemblies 10513360 and Body Assemblies 10542201 and 11731101.

3.14.2 Interchangeability - pressure retention.- The head assembly assembled to the mount and body assemblies shall be locked in position and the head assembly shall withstand an internal pressure of 5 + 0.10 pounds per square inch gage (psig) for a minimum of 2 hours without loss of pressure in excess of .2 pounds (psig).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase orer, the supplier is responsible for the performance of all inspection requirements-as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspectio requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 head assemblies plus 3 each of all items covered by SQAP. The sample shall be manufacture in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The samPle and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I. II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL .65% Defective		
101.	Collimation, pre-vibration	3.7.1.1	4.6.3
102.	Vibration	3.4	4.6.4
103.	Collimation, post-vibration	3.7.1.2	4.6.5
104.	Cleanliness	3.5	4.6.6
105.	Collimation change (due to assembling to mount)	3.7.2	4.6.7
106.	Image tilt	3.7.3	4.6.8.1
107.	Elevation accuracy	3.7.6	4.6.8.2
108.	Excursion range	3.7.4	4.6.8.3
109.	Plumb travel	3.7.5	4.6.8.4
110.	Backlash	3.7.7	4.6.8.5
111.	Resolution	3.8	4.6.9
112.	Luminous transmittance	3.9	4.6.10

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TABLE I - Continued

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
113.	Coupling travel	3.10	4.6.11
114.	Torque, input coupling	3.11	4.6.12
115.	Sealing	3.6	4.6.13
116.	Interior atmosphere	3.12	4.6.14
117.	Cleanliness	3.13	4.6.15
118.	Interchangeability - mechanical	3.14.1	4.6.16.1

MINOR: NONE

NOTE: The test in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One head assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Drawing Visual
302.	General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental. - Three head assemblies shall be selected at randoom as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.3.1	4.6.1
304.	Torque, input coupling (-65° and +150°F)	3.3.2	4.6.2
305.	Interchangeability - Pressure retention (+60° to +90°F)	3.14.2	4.6.16.2

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4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when con-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester - Two Station.

4.5.2.3.2 Orientation, pre-vibration collimation, vibration, post-vibration collimation, and collimation change due to assembling to mount.- The special testing equipment for these tests shall consist of the following main assemblies:

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a. Mounting plate assembly
b. Collimation test stand assembly
c. Vibration test stand assembly
d. Vibrating machine

4.5.2.3.2.1 Mounting plate assembly.- The mounting plate assembly utilized to mount and orient the head assembly for the pre-vibration, vibration, post-vibration, and collimation change tests shall incorporate the necessary location surfaces and head mounting features of the Mount Assembly 10513500. The mounting plate assembly shall also provide for positioning Datum Plane "A", as shown on Drawing F10542O4, in a horizontal plane within 10 seconds, and Surface "C" of the input coupling positioned 66 degrees, within 10 seconds, with respect to Datum Plane "A". Features shall be designed into the mounting plate assembly that will assure positive and accurate positioning and re-positioning (repeated removal and replacement) of the mounting plate assembly on the collimation test stand within 5 seconds in all planes.

4.5.2.3.2.2 Collimation test stand assembly. - The assembly shall function as a holder for the mounting plate assembly, with the head assembly positioned with its Datum Plane "A" in a horizontal plane. Features shall be designed into the collimation test stand assembly that will assure positive and accurate positioning and re-positioning (repeated removal and replacement) of the mounting plate assembly on the test stand assembly within 5 seconds in all planes. A target projector collimator, a colibrated penta prism, and an angle reading telescope as shown in Figure 1, shall also be part of the test stand assembly. The target projector collimator shall have an entrance pupil of at least 2.0 inches, with an appropriate crossline reticle pattern. The target projector collimator shall be positioned so that its optical axis is maintained perpendicular to Datum Plane "B" of the head assembly. The calibrated penta shall be positioned in the reflected light path of the head assembly so that its incident surface lays in a plane that is parallel to Datum Plane "A". The emergent surface of the prism shall lay in a plane that is parallel to Datum Plane "B". The angle reading telescope shall have an entrance pupil of at least 2.0 inches, and the capability of erading line of sight deviations in the "X" and "Y" coordinates in seconds or arc. The telescope shall be positioned in line with the penta prism so that its optical axis is maintained parallel to Datum Plane "A" and perpendicular to Datum Plane "B". The horizontal and vertical reticle lines of the telescope shall be superimposed on the target image lines prior to positioning the head assembly in the line of sight. The vertical reticle lines of the collimator shall be adjusted plumb. The target projector collimator and angle reading telescope shall be adjusted for infinity focus. A set-up device of the front surface reflecting mirror type shall be utilized for the alignment of the target projector collimator, penta prism, and viewing collimator. The tolerance on the incident line of sight orientation shall be 5 seconds of arc.

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4.5.2.3.2.3 Vibration test stand assembly.- The vibration test stand assembly shall function as a holder for the mounting plate assembly with the head assembly maintained in its normal operating position. Features shall be designed into the vibration test stand assembly that will assure positive and secure mounting of the mounting plate assembly on the vibration test stand. The vibration test stand shall be firmly and securely mounted on a vibration machine.

4.5.2.3.2.4 Vibrating machine.- The vibrating machine shall be capable of providing the amplitude and frequency specified in 3.4. The vibrating machine shall also be capable of vibrating the head assembly along 3 mutually perpendicular axes while the head assembly is maintained in the position specified in 3.7.

4.5.2.3.3. Accuracy requirements (orientation). - The testing equipment utilized to mount and orient the head assembly for the accuracy tests shall permit mounting Datum Plane "A", as shown of Drawing F10542204, in a horizontal plane within 10 seconds, and Surface "C" of the input coupling positioned 66 degrees within 10 seconds with respect to Datum Plane "A". The testing equipment shall also provide for articulation of Surface "C" of the input coupling to 67 degrees 30 minutes within 10 seconds, and to any angular position between its travel stops. The necessary locating surfaces and head mounting features of the Mount Assembly Drawing F1051350O shall be incorporated into the testing equipment. The testing equipment shall also provide a target projector collimator, a calibrated penta prism, and an angle reading telescope that conforms to 4.5.2.3.2.2. This testing equipment shall be positioned as specified in 4.5.203.2.2. The tolerance on the incident line of sight orientation shall be 5 seconds of arc. Unless otherwise specified, the target projector collimator and angle reading telescope shall be adjusted for infinity focus.

4.5.2.3.3.1 Plumb travel. - The testing equipment for this test shall be accurate within 5 seconds.

4.5.2.3.3.2 Image tilt. - The testing equipment for this test shall be accurate within 3 minutes.

4.5.2.3.4 Elevation accuracy, excursion range, and backlash. - The testing equipment specified in 4.5.2.3.3 shall be utilized for these tests, except the target projector collimator is not required. Auxiliary testing equipment employing an auto-collimation device, affixed to the input coupling 0f the head assembly, shall be used in place of the target projector collimator. This testing equipment shall consist of an arm assembly with a calibrated optically flat front surface mirror and an appropriate adapter that accommodates the input coupling of the head assembly positioned as shown in Figure 2 . The mirror shall be auto-collimated to the angle reading telescope when Surface "C" of the head assembly input coupling is positioned 67 degrees 30 minutes with respect to Datum Plane "A". The

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reflecting (auto-collimating) surface of the mirror shall lay in a plane that is parallel to a line passing through the vertex of the angle formed by the two side edges of the coupling, and the geometric center of rotation of the coupling. The reflecting surface of the mirror shall also remain parallel to the vertical centerline of the input coupling adapter during rotation through its full excursion of travel. The testing equipment shall be accurate within 5 seconds. The testing equipment shall be calibrated to the limits of 22° + 1° above, and 18° + 1° below the original line of sight projecting from the head assembly.

4.5.2.3.5 Set-up and calibration.- Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

NOTE 1: Caution shall be exercised during the following tests to prevent the input coupling from striking hard against the travel stops.

4.6.1 Storage temperature. - The head assembly subjected to this test shall have its shipping cover installed. The head assembly shall be pressurized with dry nitrogen to an internal pressure of 3 psi ± 0.5 psi. The nitrogen gas shall have a dew point at least as low as -25°F. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 2) to -80°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, the temperature of the test chamber shall be gradually raised to +160°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the head assembly removed from the test chamber. The shipping cover shall be removed from the head assembly. The head assembly and shipping cover shall then be subjected to the visual, tactile, and audible examination of 4.6.4.1. The head assembly and shipping cover shall show no evidence of physical failure as specified in 4.6.3 through 4.6.13 inclusive to determine overall compliance with 3.3.1. Caution shall be exercised during this test to avoid subjecting the head assembly and shipping cover to thermal shock.

NOTE 2: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

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4.6.2 Extreme operating temperature.- The head assembly subjected to this test shall have its shipping cover installed. The head assembly shall be pressurized with dry nitrogen to an internal pressure of 3 psi ± 0.5 psi. The nitrogen gas shall have a dew point at least as low as -25°F. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 2) to -65°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, and while at -65°F, the head assembly shall be subjected to the test in 4.6.12. Upon completion of this test, the temperature of the test chamber shall be gradually raised to +150°F. The head assembly shall remain at this temperature for a period of 2 hours. At the completion of this 2 hour period and while at +150°F, the head assembly shall again be subjected to the test in 4.6.12. Upon completion of this test, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) and the head assembly removed from the test chamber. The head assembly shall conform to the requirements of 3.3.2. Caution shall be exercised during this test to avoid subjecting the head assembly and shipping cover to thermal shock.

4.6.3 Collimation, pre-vibration. - Prior to the vibration test, position the head assembly as specified in 3.7, utilizing testing equipment conforming to 4.5.2.3.2.1 and 4.5.2.3.2.2. View through the eyepiece of the telescope and adjust the telescope reticle to obtain coincidence of its horizontal and vertical reticle lines with the target crossline image. Record the readings of the telescope micrometer knobs for use in the post vibration test. Subsequent to recording the pre-vibration collimation setting, remove the mounting plate assembly (with the head assembly installed) from the collimation test stand (4.5.2.3.2.2) and position it on the vibration test stand (4.5.2.3.2.3). Subject the head assembly to the test in 4.6.4. Caution must be exercised during this transfer of the mounting plate assembly. All unnecessary shocks which would tend to render the testing equipment inaccurate for subsequent tests shall be avoided.

4.6.4 Vibration. - Perform this test with the head assembly, and mount plate assembly positioned on testing equipment that conforms to 4.5.2.3.2.4. The head assembly shall be vibrated in accordance with Procedure IX, Part 1, of MIL-STD-810, except that the frequency, frequency cycling and amplitude of vibration shall be as specified in 3.4. Immediately following the vibration cycle, and prior to the examination for evidence of physical failure, the head assembly shall be subjected to the test in 4.6.5. Caution shall again be exercised during transfer of the mounting plate assembly with the head assembly. Upon completion of the test in 4.6.5, the head assembly shall be subjected to the tests specified in 4.6.4.1.

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4.6.4.1 Examination. - Subsequent to the test in 4.6.5, the head assembly shall be thoroughly examined for any evidence of physical failure. A visual and tactile examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts. An audible examination shall be made of the head assembly by shaking it and listening for loose internal parts or broken components. Any one of the possible defects defined above shall be cause to consider an assembly defective. The head assembly shall be subjected to the tests in 4.6.6 through 4.6.15 inclusive to determine overall compliance with 3.4.

4.6.5 Collimation, post-vibration.- Conduct this test subsequent to vibration and prior to the examination for physical failure resulting from vibration. Position the mounting plate assembly with the head assembly on the collimation test stand as in 4.6.3. Observe the horizontal and vertical reticle lines of the telescope with respect to the target image crosslines. Any lack of coincidence between the telescope reticle lines and the target image lines shall not exceed the tolerance specified in 3.7.1.2, as measured with the angle reading telescope. Record the readings of the telescope micrometer knobs for use in the test in 4.6.7.

4.6.6 Cleanliness. - Subsequent to the post-vibration examination (4.6.4.1), the head assembly shall be visually examined for cleanliness through the entrance window and exit aperture with the unaided eye. Inspection for condensates and grease shall be made by the technique of shadowing. Inspection for fingerprints, fractures and dirt (or foreign) particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. Cleanliness of the head assembly shall conform to the requirements of 3.5.

4.6.7 Collimation change (due to assembling to mount). - Position the head assembly on the testing equipment as specified in 4.6.3. Observe the horizontal and vertical reticle lines of the telescope with respect to the target crossline image. Obtain coincidence of the telescope reticle lines and target crossline image, if necessary. Record the readings of the telescope micrometer knobs. Remove the head assembly from the mounting plate assembly, then re-assemble the head assembly to the mounting plate, a minimum of 3 times. The coincidence of the telescope reticle lines with the target image lines shall be verified after each re-assembly of the head assembly to the mounting plate. Any deviation of the line of sight resulting from this remouval and re-assembling of the head assembly to the mounting plate shall be measured and recorded. The head assembly shall conform to the requirements of 3.7.2, as determined by the recorded values obtained at each re-assembly to the mounting plate.

4.5.8 Orientation. - The following tests shall be performed with the head assembly positioned on special testing equipment that conforms to the requirements of 4.5.2.3.3. The telescope positioned at the penta prism shall be auto-collimated to Datum Plane “A” in elevation, and the intersection of Datum Planes “A” and “B” in azimuth. Establish the optical axis

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of the target projector collimator parallel to Datum Plane "A" and perpendicular to the intersection of Datum Planes "A" and "B" by means of auto-collimation. The vertical reticle lines of the telescope and target collimator shall be plumb. Auto-collimation and alignment shall be accomplished with the head assembly removed from the line of sight.

4.6.8.1 Image tilt.- For this test, the field of view of the angle reading telescope shall be known. This value shall be multiplied by the tangent of the tilt angle specified in 3.7.3. The result of this calculation represents the maximum azimuth angle error permitted when testing the head assembly for image tilt as follows: While viewing through the angle reading telescope, position one extremity of its vertical reticle line, at the edge of the field, coincident with the corresponding extremity of the vertical target line image. The opposite extremity of the vertical target line image shall coincide with the corresponding extremity of the angle reading telescope vertical reticle line within the azimuth angle error as calculated above. The azimuth angle error shall be measured with the angle reading telescope.

4.6.8.2 Elevation accuracy.- The testing equipment specified in 4.5.2.3.3 shall be utilized for this test. Affix and orient the auxiliary testing equipment to the input coupling of the head assembly as specified in 4.5.2.3.4. With backlash removed for travel in elevation, position Surface "C" of the input coupling at 67 degrees 30 minutes with respect to Datum Plane "A". Direct a line of sight through the telescope and note the position of the reflected image with respect to the real image. Rotate Surface "C" of the input coupling so that the line of sight is deflected to at least 22 degrees above the horizontal plane, then reversing direction of rotation, deflect the line of sight to at least 18 degrees below the horizontal plane. During this rotation of the input coupling, the reflected reticle image shall not deviate from the real reticle image beyond the tolerance specified in 3.7.6, as measured on the telescope micrometer knob.

4.6.8.3 Excursion range.- Perform this test in conjunction with the test in 4.6.8.2 (elevation accuracy). Rotation of the input coupling shall conform to the requirements specified in 3.7.4.

4.6.8.4 Plumb travel. - Perform this test in sequence to the test in 4.6.8.3. Position Surface "C" of the input coupling at 67 degrees 30 minutes with respect to Datum Plane "A". While observing the intersection point of the reticle lines of the real and reflected image,rotate Surface "C" of the input coupling to the maximum elevation, then to maximum depression. During this rotation the reflected image shall not deviate in azimuth from the real image beyond the tolerance specified in 3.7.5.

4.6.8.5 Backlash. - Perform this test in sequence with 4.6.8.4. With backlash removed direction of travel, position Surface "C" of the input

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coupling 67 degrees 30 minutes with respect to Datum Plane "A". Record the position of the boresight point of the reflected image, as measured on the telescope micrometer knobs. Rotate Surface "C" of the input coupling approximately 10° in the same direction of travel, then reverse rotation of the input coupling (with backlash included) until Surface "C" of the input coupling is repositioned 67 degrees 30 minutes with respect to Datum Plane "A", without over-travel. Record this position of the boresight point of the reflected image as measured on the telescope micrometer knobs. Any difference between the recorded readings is the amount of backlash. Backlash shall not exceed the limit specified in 3.7.7.

4.6.9 Resolution.- Position the head assembly on its side with Surface "C" of the input coupling immobilized at 66 degrees with respect to Datum Plane "A". The test for resolution shall be performed utilizing an auxiliary telescope conforming to the requirements of 3.8, a resolving power chart on a wall target representing the angular subtense for the seconds of arc specified for resolution, and a supporting medium to receive this test equipment as well as the item to be tested. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-13830, and the resolving power chart shall contain 4 line sets as shown in Figure 1 therein. Resolution shall be within the value specified in 3.8, as determined by observing through the exit aperture of the head assembly with the auxiliary telescope.

4.6.10 Luminous transmittance.- Position the head assembly and Surface "C" of its input coupling as specified in 4.6.8. Diffuse luminance from a 2870° ± 100° Kelvin tungsten source shall be Positioned at the exit aperture of the head assembly. The transmittance measurement shall be made with an eye corrected photometer while scanning the entrance aperture. The transmittance shall be not less than the value specified in 3.9, as measured on the photometer. Compute the percentage of transmittance using the following:

$$\frac{\text{Luminance of source as seen through head}}{\text{Luminance of source}} \times 100\%$$

4.6.11 Coupling travel. - Rotation of the input coupling through its full excursion of travel shall be subjected to a tactile examination while observing the input coupling and the travel stop. Coupling travel shall conform to the requirements of 3.10.

4.6.12 Torque input coupling.- Standard torque measuring equipment capable 'of "measuring the torque values specified in 3.11 shall be utilized for this test. An appropriate adapter, to accommodate the input coupling, shall be used with the torque measuring equipment. Torque shall be measured while rotating the input coupling through its full range of travel. The running torque required to rotate the input coupling shall conform to the applicable requirements of 3.11 for the temperatures indicated.

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4.6.13 Sealing. - Install the shipping cover on the head assembly, and secure the poppet in Valve 10516717 in a closed position. The internal pressurizing of the head assembly to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied to the head assembly through the valve provided in the shipping cover. Pressurize the head assembly with dry nitrogen gas to the value specified in 3.6, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. Retain the applied pressure in the head assembly for a minimum of 45 minutes. Within the first 30 minutes of this 45 minute time period, the input coupling shall be rotated through a minimum of 3 full excursions (cw stop-to ccw stop-to cw stop) of its rotational range. At the end of the 45 minute time period, there shall be no evidence of leakage as indicated on the pressure gauge of the testing equipment. Upon completion of this test, release the poppet in Valve 10516717.

4.6.14 Interior atmosphere.- Perform this test in sequence to the test in 4.6.13, with the shipping covers installed. The special testing equipment specified in 4.6.13 shall be utilized in this test. A dew point tester, with an appropriate adapter to accommodate the Valve 10516717 shall be used to test the dew point of the interior atmosphere of the head assembly. The head assembly shall be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the head assembly has a dew point at least as low as minus 25°F, as measured with the dew point tester. The dew point tester shall be removed from the head assembly. The internal pressure of nitrogen. gas in the head assembly shall be brought to a measured 3 psig ± 0.5 psig. The internal pressure of the head assembly shall be monitored for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period, there shall be no evidence of leakage in the head assembly as indicated on the pressure gauge of the testing equipment. The head assembly shall conform to the requirements of 3.12. The head assembly shall then be subjected to the test in 4.6.15.

4.6.15 Cleanliness. - Perform this test in accordance with the requirements of 3.13. The accepted head assembly with shipping covers installed shall conform to the cleanliness requirements of 3.13.

4.6.16 Interchangeability.

4.6.16.1 Interchangeability - mechanical.- This test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11738174, F11738175, F11738178 and F11738179 (see 2.1). Each interchangeability gage shall be capable of being firmly secured and subsequently removed from the head assembly.

4.6.16.2 Interchangeability - pressure retention.- This test shall be performed utilizing the Fixtures, Pressure Testing conforming to Drawings

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F11738182, F11738183 and F11738184 (see 2.1). The head assembly shall be capable of being inserted into the above fixtures and secured. The Head Assembly 10513360 shall be secured to the Mount Fixture 11738184 and Body Fixture 11738182. This assembly shall be subjected to the sealing test specified in 4.6.13 for a minimum duration of 2 hours. The head assembly shall meet the requirements of 3.14.2. Repeat the above procedure with the Head Assembly 10513360 secured to Mount Test Fixture and Body Test Fixture 11738183. This pressure tetention test shall be conducted utilizing clean dry air. For purposes fo this test "Clean dry air" is that quality of air that will not deposit dirt or moisture on the components of the assembly under test when the test is conducted at a temperature between +60° and +90°F.

5. PREPAraTION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10542204. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Tank Periscope Head Assembly 10542204 is primarily intended for use as the head assembly for the gunner's periscope in the M551 Vehicle. Assembled to the Tank Periscope Body Assembly, the head assembly is a component of the unity power daylight, 9 power passive XM44E1 and XM44E2 Tank Periscopes.

6.2 Orderin data. - Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable packaging data sheet (see 5.1)
d. Provisions for First Article testing

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A204

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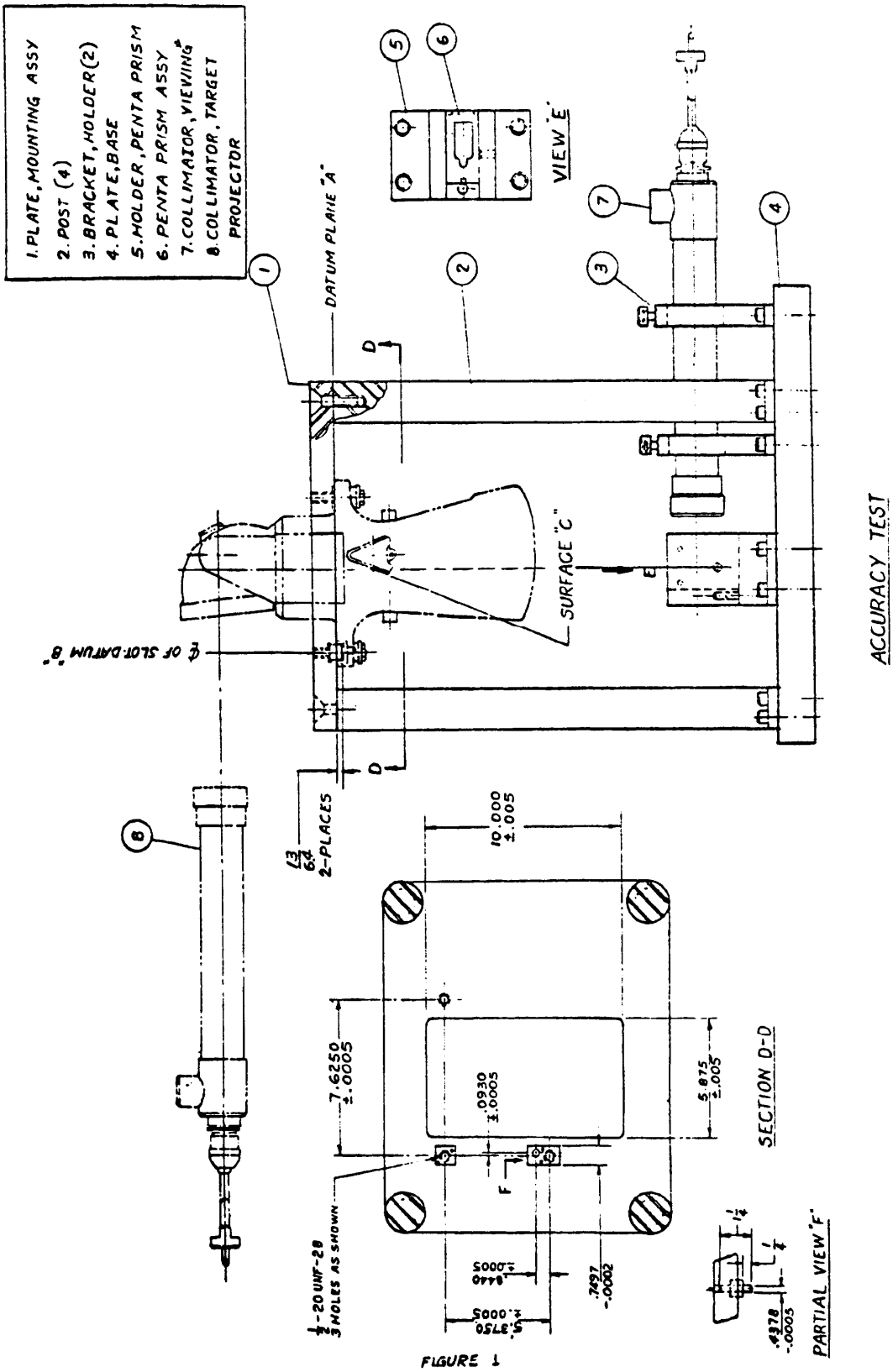


FIGURE 1

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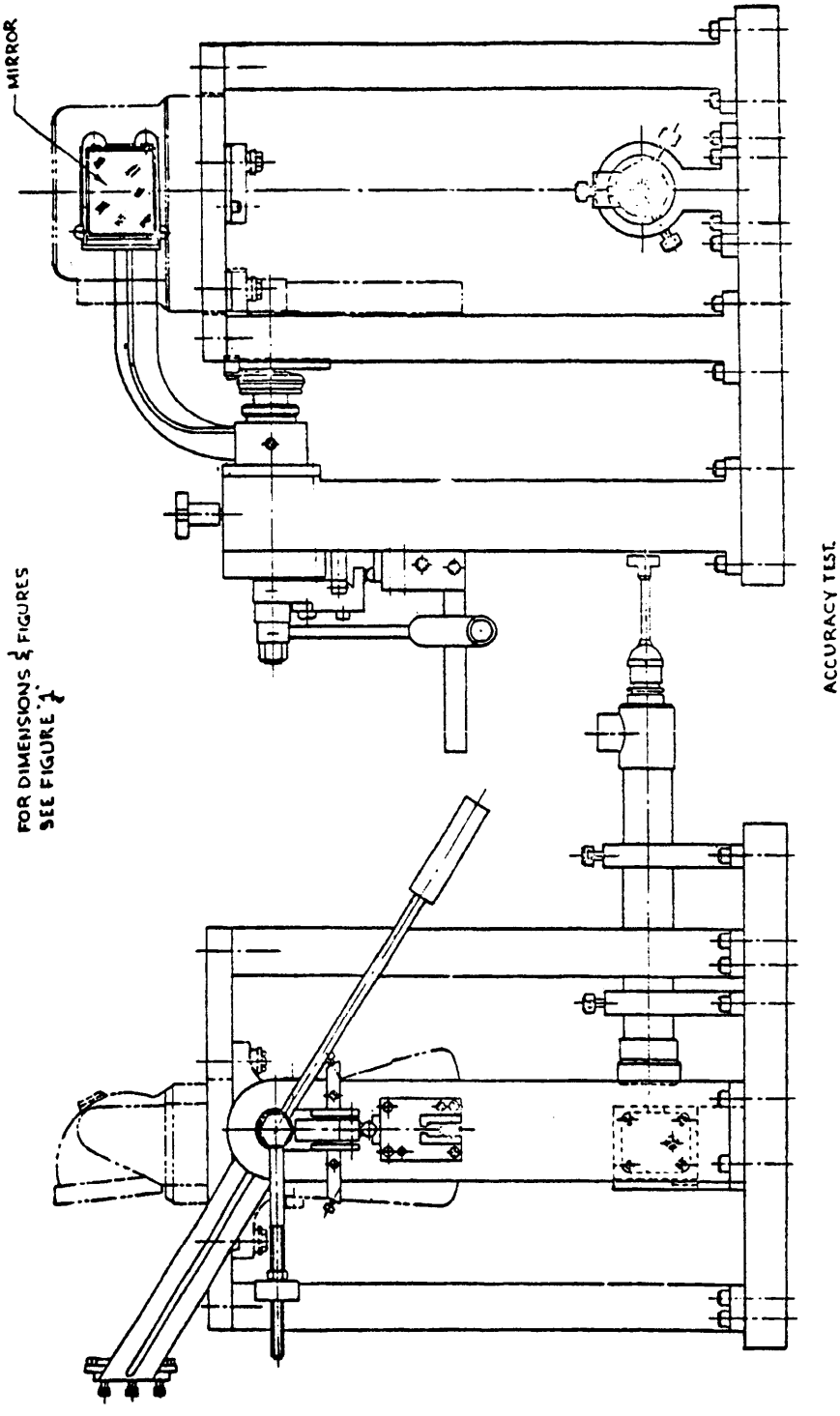


Figure 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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